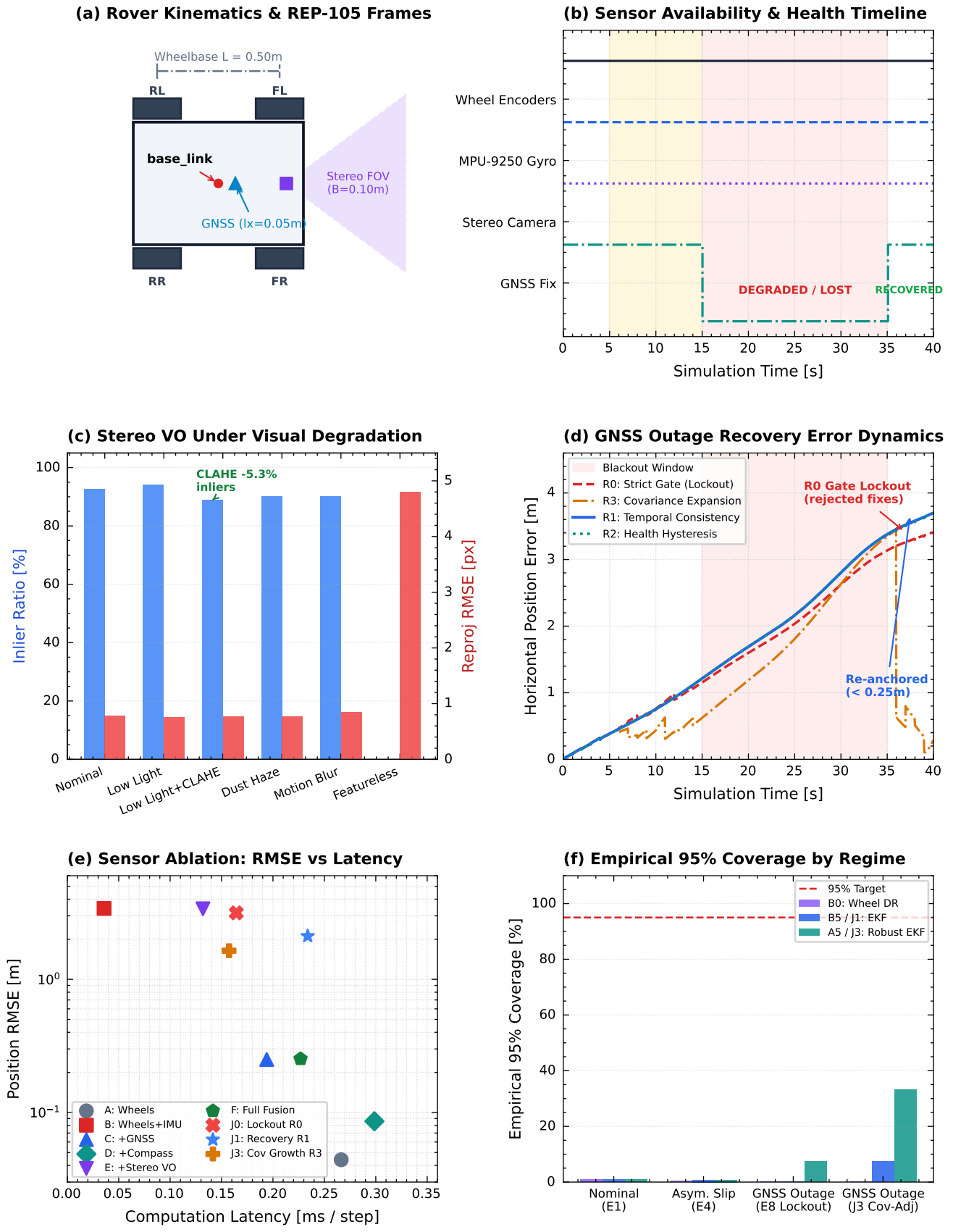
(a) Rover Kinematics & REP-105 Frames
Wheelbase L = 0.50m
RL
FL
base_link
Stereo FOV
(B=0.10m)
GNSS (lx=0.05m)
RR
FR
(b) Sensor Availability & Health Timeline
Wheel Encoders
MPU-9250 Gyro
Stereo Camera
GNSS Fix
DEGRADED / LOST
RECOVERED
Simulation Time [s]
(c) Stereo VO Under Visual Degradation
CLAHE -5.3%
inliers
Inlier Ratio [%]
Reproj RMSE [px]
Nominal
Low Light
Low Light+CLAHE
Dust Haze
Motion Blur
Featureless
(d) GNSS Outage Recovery Error Dynamics
Blackout Window
R0: Strict Gate (Lockout)
R3: Covariance Expansion
R1: Temporal Consistency
R2: Health Hysteresis
R0 Gate Lockout
(rejected fixes)
Re-anchored
(< 0.25m)
Horizontal Position Error [m]
Simulation Time [s]
(e) Sensor Ablation: RMSE vs Latency
A: Wheels
B: Wheels+IMU
C: +GNSS
D: +Compass
E: +Stereo VO
F: Full Fusion
J0: Lockout R0
J1: Recovery R1
J3: Cov Growth R3
Position RMSE [m]
Computation Latency [ms / step]
(f) Empirical 95% Coverage by Regime
95% Target
B0: Wheel DR
B5 / J1: EKF
A5 / J3: Robust EKF
Empirical 95% Coverage [%]
Nominal
(E1)
Asym. Slip
(E4)
GNSS Outage
(E8 Lockout)
GNSS Outage
(J3 Cov-Adj)